Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/05/2007
PI Name:	Levine, Benjamin D M.D.		
Project Title:	Cardiac Atrophy and Diastolic Dysfunction During and After Long Duration Spaceflight: Functional Consequences for Orthostatic Intolerance, Exercise Capacity, and Risk of Cardiac Arrhythmias		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75231-5129	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-03
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No. of PhD Candidates:		**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	In flight development phase (not yet manifested)		
Key Personnel Changes/Previous PI:	PI: Benjamin D. Levine, M.D. Co-PI: Michael W. Bungo, M.D.		
COI Name (Institution):	Bungo, Michael W. (University of Texas, Houston)		
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Task Description:	Cardiac atrophy appears to develop during spaceflight or its ground based analogues, leading to diastolic dysfunction and orthostatic hypotension. Such atrophy also may be a potential mechanism for the cardiac arrhythmias recently identified in some crew members after long duration exposure to microgravity aboard the Mir space station. Recent work by the investigators of this flight experiment has suggested that cardiac atrophy may be progressive, without a clear plateau over at least 12 weeks of bed rest, and thus may be a significant limiting factor for extended duration space missions. The purpose of this experiment is to quantify the extent and time course of cardiac atrophy and identify its mechanisms. The functional consequences of this atrophy also will be determined for cardiac filling dynamics, orthostatic tolerance, exercise tolerance, and arrhythmia susceptibility both in space on the International Space Station, and following return to earth. Hypotheses: Hypothesis 1a: Cardiac atrophy occurs in humans after prolonged spaceflight as a function of both the basal hemodynamics of microgravity (reduced wall stress and volume load) and a reduction in physical activity. Hypothesis 1b: Cardiac atrophy develops progressively during long duration spaceflight, but eventually plateaus in response to a new steady-state of cardiac work. Hypothesis 2: Cardiac atrophy during prolonged spaceflight reduces ventricular distensibility manifested as decreased end-diastolic volume for any given filling pressure. We propose that this remodeling of the heart limits the utilization of diastolic suction and impairs ventricular relaxation and filling both at rest and during orthostatic stress. This impairment of both static and dynamic diastolic function, in combination with hypovolemia is the primary mechanism leading to a reduced stroke volume during orthostasis after spaceflight. Hypothesis 3: Prolonged spaceflight leads to alterations in ventricular conduction and refractoriness, and an increase in the frequency of arrhythmias. Objectives: Objective 1a: To determine the magnitude of left and right ventricular atrophy associated with long duration space flight, and to relate this atrophy to measures of physical activity and cardiac work in-flight. Magnetic resonance imaging (MRI) will be performed pre-and post-flight as the most accurate means of measuring cardiac mass, including sufficient repetitions to document recovery. Objective 1b: To determine the time course and pattern of progression of cardiac atrophy in-flight using cardiac ultrasound. Objective 2: To determine the functional importance of cardiac atrophy for orthostatic tolerance and the regulation of stroke volume, by using a novel combination of innovative, non-invasive imaging techniques to measure the dynamic component of diastole including: a) tissue Doppler imaging to quantify ventricular wall motion with high temporal resolution; b) color Doppler M-mode echocardiography to quantify intraventricular pressure gradients and assess the magnitude of diastolic suction; and c) magnetic resonance imaging with myocardial tagging to quantify the rate of untwisting. Objective 3: To identify changes in ventricular conduction and repolarization during and after long duration spaceflight, and relate these to changes in cardiac mass. Although the direct assessment of arrhythmia risk in otherwise healthy subjects is imprecise, we will obtain hypothesis generating, pilot data using the best non-invasive tests available by measuring: 1) delayed conduction via the signal averaged ECG; 2) dispersion of refractoriness from QT dispersion; and 3) microvolt T wave alternans during routine exercise. New Information Expected: After completion of this study, the clinical expression of cardiac atrophy during long duration space flight will be defined clearly, and its significance for orthostatic tolerance and cardiac performance, both systolic and diastolic, will be elucidated. In addition preliminary information will be obtained regarding ventricular conduction and repolarization that may provide either clinical reassurance, or pathophysiolgic insight into the risk for cardiac arrhythmias. Relevance to Space and/or Earth-Based Research: Once the magnitude, time course, and inciting factors for cardiac atrophy have been determined, effective countermeasures currently being developed by the investigators in parallel ground based experiments, may be applied, focused on normalizing cardiac work and volume during long-duration spaceflight. The investigators have recently demonstrated that preventing and/or reversing the atrophy and hypovolemia during short duration head-down tilt bed rest completely eliminates orthostatic intolerance after this ground based analogue. If a similar pathophysiology can be demonstrated after long duration space flight, a strong rationale can be made for implementing the same countermeasure during missions aboard the ISS, potentially eliminating a problem, namely post-flight orthostatic hypotension, that has plagued astronauts since the inception of the manned space program. The information obtained from these space flight experiments also may be relevant for patients after prolonged confinement to bed rest, or chronic reduction in physical activity, as well as for patients with disease processes that alter cardiac stiffness such as congestive heart failure, ischemic heart disease, and normal aging.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Once the magnitude, time course, and inciting factors for cardiac atrophy have been determined, effective countermeasures currently being developed by the investigators in parallel ground based experiments, may be applied, focused on normalizing cardiac work and volume during long-duration spaceflight. The investigators have recently demonstrated that preventing and/or reversing the atrophy and hypovolemia during short duration head-down tilt bed rest completely eliminates orthostatic intolerance after this ground based analogue. If a similar pathophysiology can be demonstrated after long duration space flight, a strong rationale can be made for implementing the same countermeasure during missions aboard the ISS, potentially eliminating a problem, namely post-flight orthostatic hypotension, that has plagued astronauts since the inception of the manned space program. The information obtained from these space flight experiments also may be relevant for patients after prolonged confinement to bed rest, or chronic reduction in physical activity, as well as for patients with disease processes that alter cardiac stiffness such as congestive heart failure, ischemic heart disease, and normal aging.

Task Progress:	In the past year, we worked with our NASA team to bring experiment E377 to the Prelminary Design Review. This work was completed and the PDR scheduled for February 2006. However 3 days before the PDR, it was put on hold while NASA administration worked through the decision process on how to move their research effort in cardiovascular issues forward. Over the past few months, we have made quite dramatic progress. A new cardiovascular discipline team has been created including the PI teams, Flight Medicine, and JSC CV laboratory. We have decided to move the project forward focusing on two key aims which fill NASA defined "gaps": 1). defining the magnitude and time course of cardiac atrophy using ground based MRI, and flight based echo. The inflight echo will be performed with remote guidance using the ADUM platform; 2). defining the presence or absence of changes in electrophysiological substrate as a direct consequence of spaceflight. This aim will be accomplished primarily from using a "new" Holter system that will have sufficient temporal resolution to allow analysis for late potentials (signal averaging) and T wave alternans, as well as QT variability. The latter analysis will be done in collaboration with the investigators of the NASA Holter SMO.
Bibliography Type:	Description: (Last Updated: 03/19/2026)
Articles in Peer-reviewed Journals	Popovic ZB, Prasad A, Garcia MJ, Arbab-Zadeh A, Borowski A, Dijk E, Greenberg NL, Levine BD, Thomas JD. "Relationship among diastolic intraventricular pressure gradients, relaxation, and preload: impact of age and fitness." Am J Physiol Heart Circ Physiol. 2006 Apr;290(4):H1454-9. PMID: 16284230 , Apr-2006
Articles in Peer-reviewed Journals	Mitsis GD, Zhang R, Levine BD, Marmarelis VZ. "Cerebral hemodynamics during orthostatic stress assessed by nonlinear modeling." J Appl Physiol. 2006 Jul;101(1):354-66. Epub 2006 Mar 2. PMID: 16514006 , Jul-2006
Articles in Peer-reviewed Journals	Shibata S, Zhang R, Hastings J, Fu Q, Okazaki K, Iwasaki K, Levine BD. "Cascade model of ventricular-arterial coupling and arterial-cardiac baroreflex function for cardiovascular variability in humans." Am J Physiol Heart Circ Physiol. 2006 Nov;291(5):H2142-51. Epub 2006 Jun 9. PMID: 16766646 , Nov-2006
Articles in Peer-reviewed Journals	Fu Q, Shook RP, Okazaki K, Hastings JL, Shibata S, Conner CL, Palmer MD, Levine BD. "Vasomotor sympathetic neural control is maintained during sustained upright posture in humans." J Physiol. 2006 Dec 1;577(Pt 2):679-87. Epub 2006 Sep 28. PMID: 17008377 , Dec-2006
Articles in Peer-reviewed Journals	Claassen JA, Zhang R, Fu Q, Witkowski S, Levine BD. "Transcranial Doppler estimation of cerebral blood flow and cerebrovascular conductance during modified rebreathing." J Appl Physiol. 2006 Nov 16; [Epub ahead of print] PMID: 17110510 , Nov-2006
Articles in Peer-reviewed Journals	Iwasaki KI, Levine BD, Zhang R, Zuckerman JH, Pawelczyk JA, Diedrich A, Ertl AC, Cox JF, Giller CA, Ray CA, Lane LD, Buckey JC, Baisch F, Eckberg DL, Robertson D, Biaggioni I, Blomqvist CG. "Human cerebral autoregulation before, during, and after spaceflight" J Physiol. 2006 Dec 21; [Epub ahead of print] PMID: 17185344 , Dec-2006